The benefit of daily photoprotection

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Background: It is now recognized that both ultraviolet (UV)-A and UVB wavelengths participate in the generation of photodamaged human skin during sun exposure. During usual daily activities, an appropriate protection against solar UV exposure should prevent clinical, cellular, and molecular changes potentially leading to photoaging.

Objective: This study was designed to evaluate in human beings the protection afforded by a day cream containing a photostable combination of UVB and UVA filters and thus protect against the UV-induced skin alterations.

Results: In solar-simulated radiation exposed and unprotected skin sites we observed melanization. The epidermis revealed a significant increase in stratum corneum and stratum granulosum thickness. In the dermis, an enhanced expression of tenascin and a reduced expression of type I procollagen were evidenced just below the dermoepidermal junction. Although no change in elastic fibers in exposed buttock skin was seen, a slightly increased deposit of lysozyme and alpha-1 antitrypsin on elastin fibers was observed using immunofluorescence techniques. A day cream with photoprotection properties was shown to prevent all of the above-described alterations.

Limitations: This study was performed on a limited number of patients (n = 12) with specific characteristics (20-35 years old and skin type II and III). Two dermal alterations were evaluated by visual assessment and not by computer-assisted image analysis quantification.

Conclusion: Our in vivo results demonstrate the benefits of daily photoprotection using a day cream containing appropriate broad-spectrum sunscreens, which prevent solar UV-induced skin damages. (J Am Acad Dermatol 2008;58:S160-6.)

During usual daily activities, an appropriate protection against solar ultraviolet (UV) exposure should prevent clinical, cellular, and molecular changes potentially leading to photoaging.[1-3] In body sites regularly exposed to sun, skin damage caused by photoaging is superimposed on tissue degeneration resulting from chronologic aging. Chronologic aging alone, as seen in sun-protected sites, leads to remarkably few clinically apparent changes. In contrast, photoaging is associated with marked cutaneous alterations clinically characterized by wrinkles, roughness, sallowness, mottled pigmentation, telangiectasia, and a variety of benign and malignant neoplasms.[1,3] Histologic and ultrastructural studies have shown that changes in photoaged skin were mainly found in dermal connective tissue.[3-5] The extracellular matrix of the dermis is primarily composed of type I collagen, associated with a lower amount of type III collagen, elastin, proteoglycans, and fibronectin. Sun-exposed sites usually display a loss of mature type I collagen and an increase in collagen III/I ratio. A significant correlation has been found between reduced level of

Abbreviations used:

DC:	day cream
DEJ:	dermoepidermal junction
ITA:	individual typologic angle
MED:	minimal erythemal dose
PF:	protection factor
SC:	stratum corneum
SG:	stratum granulosum
SSR:	solar-simulated radiation
UV:	ultraviolet

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type I collagen and the severity of photodamage in human skin.[6] The histopathologic hallmark of photoaging is a massive accumulation of elastotic material in the upper and mid dermis.[3-7] This process, known as solar elastosis, has been ascribed to changes in elastin, the main component of elastic fibers, because the accumulated material was identified with elastin-specific histologic stains. The accumulation of elastotic material is accompanied by degeneration of the surrounding collagen meshwork.[8]

This study was designed to evaluate the efficacy of a day cream (DC) containing a photostable combination of UV absorbers, providing a broad-spectrum protection. Molecular, functional, and structural cutaneous changes induced by a 6-week exposure to low doses of solar-simulated radiation (SSR) in young volunteers were used for this evaluation.

METHODS

Study population

Healthy Caucasian volunteers were enrolled in the study after signing an informed consent that was approved by a medical ethics committee. The untanned upper buttock region was used as the site for exposure.

UV source and dosimetry

UV SSR source used was an Oriel 1000-W xenon arc solar simulator (Oriel Corporation, Stratford, Conn) equipped with a Schott WG 320/2-mm filter (Schott, Clichy, France) and two step-plate dichroic mirrors to deliver a UV solar-simulating spectrum. A 2 mm–thick filter was chosen because it cuts off more UVB radiations (UVB/UVA = 13) than usual WG320 filters used to simulate zenithal UV sun spectrum (UVB/UVA = 11). The spectral power distribution at the skin level (40 cm from the lamp) was measured with a calibrated spectroradiometer (Macam 3010, Scotland) (Fig 1). The daily output was monitored with a UV radiometer (Centra Osram, Germany) equipped with UVA and UVB sensors.

Exposure regimen

Four areas (3 × 3 cm) were randomized and delineated with a template on the buttock of each volunteer. One area received no exposure and no product (recorded as control). The 3 other areas were exposed for 5 d/wk for 6 weeks to one individual minimal erythemal dose (MED) of SSR per exposure. One area received no product (+SSR) before exposure, another one was pretreated with vehicle of oil/water emulsion (+SSR/+Veh), and the third one was pretreated with the DC with photoprotection containing butyl methoxydibenzoylmethane avobenzone (1.5%) (Parsol 1789), a long UVA

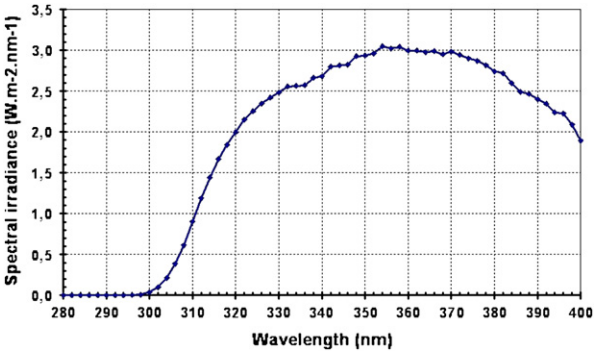


Fig 1. Spectral power distribution at skin level of radiations emitted by filtered xenon arc solar simulator. Spectrum was measured with calibrated spectroradiometer. Relative irradiance is expressed in W. m-2. nm-1.

absorber (Givaudan-Roure Vernier, Switzerland), terephtalylidene dicamphor sulfonic acid (1.5%, Mexoryl SX, USAN: ecamsule),[9] a broad-spectrum UVA absorber (Chimex SA, France), and octocrylene (4%) (Uvinul 539), a UVB absorber (BASF, Germany) (+SSR/+DC). The DC or vehicle was applied 30 minutes before each exposure at 2 mg/cm^2 on the 3- × 3-cm randomized area, which corresponds to 9 cm^2, so 18 mg (2 mg/cm^2) were applied on this area. Each product was placed in a syringe and the 18 mg necessary were weighed and applied on the specified area. The DC formulation was characterized by its sun-protection factor (PF), UVA PF, and absorption curve. UVA PF was determined by using an in vivo method based on the determination of persistent pigment darkening dose and the absorption curve was obtained using the method of Diffey and Robson,[10] which measured UV radiation transmitted through a substrate (Transpore tape, 3M Company), with and without product applied (2 mg/cm^2), using a spectroradiometer, as recommended by the American Academy of Dermatology.[11]

Skin erythema and pigmentation

Skin erythema and pigmentation were assessed using a chromameter (CR200, Minolta, Japan) with an illuminating C system as described previously.[12] This technique was used to objectively follow the dynamics and kinetics of tanning induced by our multiexposures regimen. The output was given in terms of L*, a*, and b* chromaticity coordinates (means of quadruplicate measurements). The specific directions in the 3-dimensional L*, a*, and b* color space of the tanning components, ie, erythema, immediate pigmentation, and constitutional melanization, and the resulting tanning pathways, may be analyzed and defined in the 3-dimensional color space using a vectorial method.[13] Individual

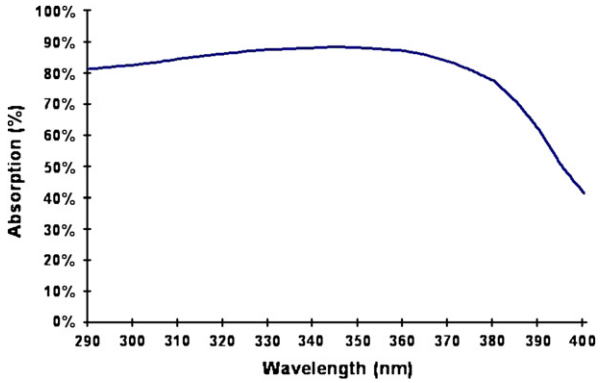


Fig 2. Absorption curve of cream for daily use (DUC). Absorption spectra of DUC was obtained spectroradiometrically between 290 and 400 nm by applying 2 mg/cm^2 of product on Transpore tape (3M Company).

typologic angle (ITA) (ITA degrees = arc tangent[(L* −50)/b*] were calculated. ITA degree characterizes the current degree of skin melanization (basic or acquired). Ranges of ITA degree values allow defining skin color categories ("very light" with ITA degree values > 55 to "black" with ITA degrees < −10). For all these parameters, the results were expressed as the variation between day 43 and day 1.

Histology and immunohistochemistry

Biopsies were performed under local anesthesia 72 hours after the last exposure. For histologic studies, skin samples were fixed in 10% buffered formalin, embedded in paraffin, and processed for light microscopy. Hematoxylin-phloxin-saffron was used for overall morphologic evaluation by a pathologist. Melanin content was quantified by visual assessment and a score summing up the individual grades was calculated and varied from 0 (no visible melanin) to 5. Grading scale corresponded to 0.5 for minimal level of melanin in the basal layer of the epidermis and additional 0.5 for increased melanin content in this layer.

The detailed melanin grading system is the following: 0 = no visible melanin; +0.5 = minimal melanin level in the stratum basal; +0.5 = increased melanin content in the stratum basal; +0.5 = minimal melanin level in the stratum spinosum; +0.5 = increased melanin content in the stratum spinosum; +0.5 = minimal melanin level in the stratum granulosum (SG); +0.5 = increased melanin content in the SG; +0.5 = minimal melanin level in the stratum corneum (SC); +0.5 = increased melanin content in the SC; +0.5 = minimal melanin level in the dermis; and +0.5 = increased melanin content in the dermis.

The same graduation was used for each epidermal layer (stratum spinosum, SG, and SC) and for melanin in the dermis. All the measurements were performed in a blind fashion.

For indirect immunofluorescence microscopy, cryostat-cut sections were air dried, rinsed in phosphate-buffered saline pH 7.2 (Biomerieux Laboratories, France), and immunolabeled at room temperature with specific antisera diluted in phosphate-buffered saline as previously described.[12] In negative controls, the primary antibody was omitted or nonimmune IgG was used instead. Intensity and area of immunostaining were evaluated either by a computer-assisted image analysis system or by visual assessment. The same operator analyzed all samples. The results obtained with the image analysis system were expressed in arbitrary units taking into account information about the number of pixels within the measurement frame and the level of illumination they represent (256 discrete gray levels). For visual evaluation (lysozyme and alpha-1 antitrypsin deposit) we used a 3-point grading scale with a 0.5 increment related to the increase of the intensity and/or the repartition of the deposition (0 corresponding to normal deposition).

Statistical analysis

All results are expressed as means ± SD. Statistical analysis was based on a mixed model of variance analysis, including volunteer as random factor and treatment as fixed factor (SAS, Proc Mixed), which was followed by Dunnett test to compare each exposed area with the control. The other comparisons (+SSR vs +SSR/+Veh, +SSR vs SSR/+DC, and +SSR/+Veh vs +SSR/+DC) were performed using corresponding contrasts. All comparisons were made at a two-tailed significant level of 5%.

RESULTS

The 12 volunteers (6 male and 6 female) enrolled in this study were 20 to 35 years old with skin types II or III. Individual MEDs was determined before exposure or suntanning on each volunteer and averaged 0.86 J/cm^2 UVB and 11 J/cm^2 UVA (spectroradiometrically measured).

UV absorption properties of the DC

The sun-PF of the DC with photoprotection was 8.39 ± 1.22 and the UVA PF value, determined using the persistent pigment darkening method, was 7.4 ± 2.4. The well-balanced protection in the UVB and UVA range is clearly illustrated by the shape of the absorption curve (Fig 2).

Skin pigmentation

Daily irradiation with 1 MED of SSR induced a significant increase in the skin pigmentation

Table I. Chromometry

Parameters (n = 12)	Control	+SSR	+SSR/+veh	+SSR/+DC
ΔL*§	−0.77 ± 1.97	−15.82 ± 2.69*	−15.82 ± 3.76*	−2.98 ± 2.02*†‡
Δb*§	0.60 ± 1.15	5.67 ± 2.10*	6.02 ± 1.63*	1.33 ± 0.86†‡
ΔITA degrees§	−1.90 ± 2.34	−39.02 ± 7.65*	−39.29 ± 7.85*	−6.30 ± 3.00†‡
Δa*§	0.35 ± 2.07	6.37 ± 2.02*	6.20 ± 2.70*	1.51 ± 2.05†‡

Data are means ± SD.
ITA, Individual typologic angle; *+SSR*, solar-simulated radiation (SSR) without pretreatment; *+SSR/+DC*, SSR and pretreatment with day cream; *+SSR/+veh*, SSR and pretreatment with vehicle of oil/water emulsion.
*$P \leq .05$ compared with control site.
†$P \leq .05$ compared with +SSR/+veh site.
‡$P \leq .05$ compared with +SSR site.
§Day 43-day 1 values.

Table II. Morphometry of the epidermis

Parameters (n = 12)	Control	+SSR	+SSR/+veh	+SSR/+DC
No. of stratum corneum layers	14.5 ± 2.3	20.6 ± 4.2*	19.8 ± 3.6*	15.0 ± 1.4†‡
No. of stratum granulosum layers	1.0 ± 0.0	2.1 ± 0.6*	1.8 ± 0.7*	1.1 ± 0.2†‡
Viable epidermis thickness (μm)	79.7 ± 12.1	88.8 ± 5.8*	85.2 ± 8.8	90.0 ± 12.0*
Melanization (visual score)	1.3 ± 0.5	3.5 ± 1.0*	3.2 ± 0.7*	1.4 ± 0.7†‡

Data are means ± SD.
+SSR, Solar-simulated radiation (SSR) without pretreatment; *+SSR/+DC*, SSR and pretreatment with day cream; *+SSR/+veh*, SSR and pretreatment with vehicle of oil/water emulsion.
*$P \leq .05$ compared with control site.
†$P \leq .05$ compared with +SSR/+veh site.
‡$P \leq .05$ compared with +SSR site.

(Table I), which increased similarly in the +SSR and the +SSR/+Veh sites throughout the study. The measurements showed that ITA degrees decreased steadily throughout the study. From light color (55 degrees > ITA degrees > 41 degrees), the SSR-exposed areas turned to matte (28 degrees > ITA degrees > 10 degrees). In the control and +SSR/+DC sites, no pigmentation change was observed during the 6-week treatment. Histologic examination of melanization after Fontana-Masson staining of skin sections at day 43 confirmed these observations (Table II).

Histology and immunohistochemistry

All the measurements were performed in a blind manner. Morphometric examination of the epidermis after SSR exposure showed a significant thickening of the SC, SG, and viable epidermis (including stratum basal, stratum spinosum, and SG). The two first changes were significantly prevented by pretreatment with the photoprotecting DC (+SSR/+DC) (Table II). The thickening of viable epidermis observed in the DC-pretreated site was not because of the mechanical effect of massage (not shown for the vehicle pretreatment) but perhaps because of a slight proliferative effect of one of the UV absorbers contained in the DC.[14]

No dermal infiltrates were observed after 6-week SSR exposure, neither at a histologic level nor by visual assessment of the immunostained pattern obtained using antisera directed against human vimentin or factor XIII-A (data not shown). Various matrix components were investigated by immunohistochemistry. Tenascin, a large glycoprotein, minimally expressed just below the dermoepidermal junction (DEJ) in normal-appearing skin,[2,15] was significantly increased in +SSR site but remained clearly localized just below the DEJ. Types I and III procollagens were predominantly located in the extracellular papillary dermis.[2] In +SSR biopsy specimens, type I procollagen decreased significantly whereas type III was unaltered. Pretreatment with the photoprotecting DC (+SSR/+DC) prevented type I procollagen decrease. Using a double staining procedure, we looked simultaneously at the elastin component of elastic fibers and lysozyme or alpha-1 antitrypsin, which have been described to be associated with damaged elastic fibers.[2,16,17] Six-week SSR exposure (+SSR) did not induce any significant change in the elastin network. However, lysozyme and alpha-1 antitrypsin deposits on elastin fibers were significantly increased in +SSR skin biopsy specimens. These modifications, considered as early events of solar elastosis, were prevented

Table III. Dermal alterations

Parameters (n = 12)	Control	+SSR	+SSR/+veh	+SSR/+DC
Tenascin, AU	4.9 ± 3.4	8.2 ± 4.9*	8.9 ± 3.4*	4.8 ± 2.8†‡
Type I procollagen, AU	7.7 ± 3.4	5.3 ± 3.9*	6.8 ± 4.2	8.7 ± 4.5‡
Type III procollagen, AU	2.8 ± 0.9	3.4 ± 1.9	3.6 ± 1.0	3.4 ± 0.8
Elastin, AU	22.5 ± 5.7	21.6 ± 3.0	20.6 ± 5.3	20.8 ± 4.5
Lysozyme deposit (visual assessment)	0.1 ± 0.2	0.8 ± 0.5*	0.8 ± 0.3*	0.4 ± 0.5†‡
α-1 Antitrypsin deposit (visual assessment)	1.0 ± 0.6	1.8 ± 0.7*	1.6 ± 0.5*	1.2 ± 0.6

Data are means ± SD.
AU, Arbitrary units; *+SSR*, Solar-simulated radiation (SSR) without pretreatment; *+SSR/+DC*, SSR and pretreatment with day cream; *+SSR/+veh*, SSR and pretreatment with vehicle of oil/water emulsion.
*$P \leq .05$ compared with control site.
†$P \leq .05$ compared with +SSR/+veh site.
‡$P \leq .05$ compared with +SSR site.

by pretreatment with the photoprotective cream (+SSR/+DC) (Table III).

DISCUSSION

Repeated solar exposure is the major environmental insult that contributes to clinical and histologic changes in sun-exposed skin (eg, face, hands). The effects of repeated suberythemal solar UV exposure on human skin have been poorly investigated.[18,19] Because UV exposure acting on 7-dehydroxycholesterol conduct to the production of vitamin D, which may be part of an intrinsic protective mechanism against UV damage,[20] such exposure has almost certainly important long-term consequences such as accelerated skin aging. The dermal damage induced by repeated UV irradiation is mainly reflected histologically by the disorganization of collagen fibrils,[21] and accumulation of abnormal elastin-containing material.[3,8] These alterations are involved in the wrinkle formation in sun-exposed skin. Biochemical evidence of connective-tissue alterations in photoaged skin includes reduced levels of type I and III collagen precursors,[2,22] an increased number of cross-links,[23] type III/I collagen ratio,[24] and elastin content. A significant correlation was found between the reduced level of collagen type I and the severity of photodamage in human skin.[6] Our immunohistologic findings evidenced a decrease in type I procollagen, induced by our UV regimen, principally located at the DEJ.[2] This site has been reported to be the location of collagen neosynthesis.[25]

The histologic hallmark of dermal photoaging is elastosis, a quantitative and qualitative change affecting elastic fibers.[3] Accumulation of elastic fibers is dependent on the degree of sun exposure.[26] Using immunohistochemical techniques, we and other investigators detected increasing lysozyme and alpha-1 antitrypsin deposits on photodamaged elastic fibers, with a staining intensity positively correlated with the degree of sun damage.[16,27,28] The components of elastic fibers to which lysozyme or alpha-1 antitrypsin are bound are not yet known, but they seem to be distinct (staining just below the DEJ for alpha-1 antitrypsin).[28] Because lysozyme or alpha-1 antitrypsin at high concentrations inhibit the activity of collagenase and elastase,[29] these proteins could protect elastic fiber components from proteolysis.[30] Deposits of lysozyme or alpha-1 antitrypsin were enhanced after repeated SSR exposure.[2] Because we were unable to detect any histologic alteration of the elastic fibers network in SSR-exposed sites, an increase in these deposits may be viewed as an early marker for the UV radiation-induced elastotic process.[28]

It is important to note that under any meteorologic condition, the UVA irradiance is at least 17 times higher than the UVB irradiance and that UVA rays may be received while UVB are absent (UVA pass through glass, which absorbs UVB[31]). In our SSR radiation source, to better simulate zenithal sun spectrum (UVA/UVB = 17), we used a 2 mm—thick filter that cuts off more UVB (UVB/UVA = 13) than the 1 mm—thick filter most commonly used (UVA/UVB = 11). For most of biological effects, UVA is far less active than UVB, nevertheless, it has been previously evidenced that repetitive UVA exposure induces damage as evidenced by the histologic and immunohistologic endpoints assessed in this study.[12,17] In addition, about two thirds of cumulative erythemal UV dose received per year (in the United States, this dose is around 105 MED, for a skin type II) is accumulated inadvertently, when no topical protection is used.[32,33] Based on these data it has been recommended that life-long protection against excess UV radiation should be used.[34]

In conclusion, using repeated low doses of SSR for 6 weeks allows the production of some of the major alterations observed and/or participating in the long-term photoaging process (eg, reduced level of type I

collagen precursor, increased lysozyme deposit on elastic fibers) even if a strong induced pigmentation arose, conferring a progressive skin autoprotection. This protection seemed apparently not so effective because we noticed in the SSR-exposed sites, despite the strong pigmentation, typical alterations of photodamage. This regimen has been used to demonstrate the efficacy of a DC with photoprotection containing a photostable combination of broad-spectrum absorbers. This DC prevented some biological end points as described in this study, which correspond to epidermal and dermal targets of UV radiation. All these data support the rationale for a relevant daily photoprotection.

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